QUANTITATIVE METHODS 1

Worksheet 2 – Measures of central tendency

Detailed solutions

Exercise 1 – Detailed solution

A. Mode, median and mean

$$Mo = 20, \qquad Me = \frac{20+22}{2} = 21,$$

$$\bar{x} = \frac{18+20+20+22+25+95}{6} = 33.33.$$

The median better represents a typical customer because 95 is extreme. The mean is still useful for revenue because $n\bar{x}$ reproduces the total basket value.

B. Weighted mean

$$\bar{x}_w = 14(0.30) + 11(0.30) + 16(0.40) = 13.90.$$

C. Geometric mean

$$1 + g = \sqrt[3]{1.05 \times 0.98 \times 1.12} \approx 1.048442,$$

$$g \approx 4.84\%.$$

D. Harmonic mean

$$H = \frac{2}{\frac{1}{60} + \frac{1}{90}} = 72 \text{ km/h}.$$

Exercise 2 – Detailed solution

The count densities are $1, 3, 4, 8, 12, 7, 5, 1, 1$, so the modal class is $[30, 35[$.

$$Me = 30 + 5\frac{105 - 80}{60} \approx 32.08.$$

$$Q_1 = 26 + 4\frac{52.5 - 48}{32} \approx 26.56,$$

$$Q_3 = 35 + 5\frac{157.5 - 140}{35} = 37.50.$$

Class midpoints are $12, 16, 22, 28, 32.5, 37.5, 42.5, 47.5, 52.5$ and

$$\sum n_i c_i = 6665.$$

Hence:

$$\bar{x} = \frac{6665}{210} \approx 31.74.$$

For 60% to afford at least price p, 40% must lie below it:

$$0.40(210) = 84,$$

$$p = 30 + 5\frac{84 - 80}{60} \approx 30.33 \text{ euros}.$$

Exercise 3 – Solution

Situation	Preferred indicator
Most frequently sold category	Mode.
Typical salary with a few extreme high salaries	Median.
Average price with different quantities	Weighted mean.
Average growth over several years	Geometric mean.
Average speed over equal distances	Harmonic mean.
Symmetric numerical distribution without extremes	Arithmetic mean.